

2018 National Common Entrance Examination (Past Questions and Answers)

Paper 1

Part A: Mathematics and General Science

National Common Entrance
Examination 2018

Solutions to Mathematics

1.

$$\begin{array}{c} 60s \rightarrow 1min \\ \swarrow 8min \\ = \frac{8min \times 60sec}{1min} = 480sec \end{array}$$

$$\therefore 480seconds + 12seconds \\ = 492seconds$$

2.

$$\begin{aligned} & \left(\frac{2}{5} + \frac{2}{7} \right) * \left(1 - \frac{1}{8} \right) \\ \Rightarrow & \left(\frac{14+10}{35} \right) * \left(\frac{1}{1} - \frac{1}{8} \right) \\ \Rightarrow & \left(\frac{24}{35} \right) * \left(\frac{8-1}{8} \right) \\ \Rightarrow & \left(\frac{24}{35} \right) * \left(\frac{7}{8} \right) \\ = & \frac{3 \cancel{24}}{\cancel{28} 5} \times \frac{7}{8} = \underline{\underline{\frac{3}{5}}} \end{aligned}$$

3.

The total students in the college is 750 = 100%
If 70% are girls, then 30% are boys.

i.e. 30% of 750 are boys

$$\text{i.e. } \frac{30}{100} \times \frac{750}{1} = 225$$

4.

$$\begin{aligned} \sqrt{5\frac{4}{9}} &= \sqrt{\frac{49}{9}} = \frac{\sqrt{49}}{\sqrt{9}} = \frac{7}{3} \\ &= \underline{\underline{2\frac{1}{3}}} \end{aligned}$$

5.

$$\begin{array}{r} 2001 - 100 \\ 2004 - 250 \end{array}$$

$$\begin{aligned} \therefore \text{Ratio} &\Rightarrow 100:250 \\ &\Rightarrow \frac{100}{250} = \frac{2}{5} \end{aligned}$$

$$\therefore \text{Ratio} = \underline{\underline{2:5}}$$

6.

Length: Breadth
7:5

$$x:10$$

$$\Rightarrow \frac{7}{5} = \frac{x}{10} \Rightarrow 5x = 70$$

$$\therefore x = \frac{70}{5} = \underline{\underline{14m}}$$

$\therefore$ length of room = 14m.

7. $10\% \text{ of } 80 \Rightarrow \frac{10}{100} \times \frac{80}{1} = 8.$

$\therefore \text{Increase } 80 \text{ by } 10\% = 80 + 8$
 $= \underline{88}.$

8. 12 cartons of biscuit = ₦3,000
 1 carton of biscuit = $\frac{₦3,000}{12}$
 $= \underline{₦250.00}$

$\therefore 7 \text{ carton of biscuit} = 7 \times ₦250$
 $= \underline{₦1,750.00}$

9. $\frac{36}{100} = \frac{18}{50} = \frac{9}{25}$

10. ₦5,000 — 10 days
 ? — 1 day
 $\Rightarrow \frac{1 \text{ day} \times ₦5,000}{10 \text{ days}} = \frac{5,000}{10} = ₦500$

$\therefore \text{The man earns } ₦500 \text{ per day}$
 In 14 days, the man will earn 500×14
 $= \underline{₦7,000}.$

11. Simple Interest, $I = \frac{PRT}{100} = \frac{600 \times 5 \times 8}{100}$

$\therefore I = 6 \times 5 \times 8 = \underline{₦240.0}$

12. 400g : 1Kg.

Recall that 1000g = 1Kg

$\therefore 400g : 1000g$

$\Rightarrow \frac{400g}{1000g} = \frac{4}{10} = \frac{2}{5}$

$\Rightarrow \underline{2:5}.$

13.

8 children eat in 12 days

1 child will eat in $\frac{12}{8}$ days

6 children will eat in $\frac{12}{8} \times 6 = 9$
 days

$\therefore 6 \text{ children will eat in } 9 \text{ days}$

14.

$2x - x + 4x = x + 4x = 5x$

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